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Deutsche Telekom Chair of Communication Networks

First AirMetro Workshop

T9 – Communication Networks

Network Handling and Health for Reliable
Communication in AAM

Supervisor - Prof Frank Fitzek

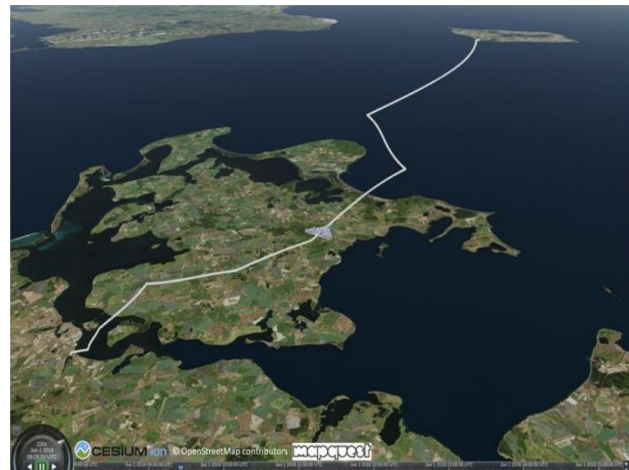
Dresden, November 22, 2024



Where are we ...

Hybrid satellite and 5G network architecture shaping the future of drone communication

Drones require reliable data networks to perform operations such as command and control or autonomous flying. EU-funded researchers demonstrated a hybrid architecture combining cellular and satellite networks that enables fast and safe flying of commercial drones in urban areas.



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[1]



[2]

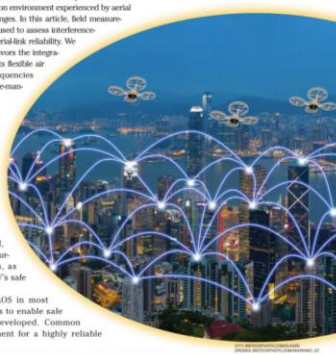
ENABLING CELLULAR COMMUNICATION FOR AERIAL VEHICLES

Providing Reliability for Future Applications

Due to safety concerns, a reliable radio communication link is a key component in the future application of unmanned aerial vehicles (UAVs), as it will enable beyond visual line-of-sight (BVLOS) operations. In terms of cost and deployment time, radio communication for aerial vehicles will greatly benefit from the ready-to-market infrastructure and ubiquitous coverage of cellular networks. However, these are optimized for terrestrial users, and the different propagation environment experienced by aerial vehicles poses some interference challenges. In this article, field measurements and system-level simulations are used to assess interference-mitigation solutions that can improve aerial-link reliability. We then discuss how 5G New Radio (NR) leverages the integration of UAVs into cellular networks, as its flexible air interface and beamforming-suited frequencies facilitate the deployment of interference-mitigation solutions.

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Background
In recent years, the popularity of UAVs (also known as drones) has experienced rapid growth as technological improvements have significantly reduced their cost and size. UAVs present great potential to reduce the risk, cost, and deployment time of many commercial and civil applications such as surveillance and monitoring, agriculture inspection, and disaster relief, among others [1]. Such potential is currently limited by early regulations, as public authorities need to ensure UAV's safe integration into the manned airspace. Flight ranges are limited to VLOS in most countries, while technical solutions to enable safe BVLOS flight ranges are being developed. Common to these solutions is the requirement for a highly reliable



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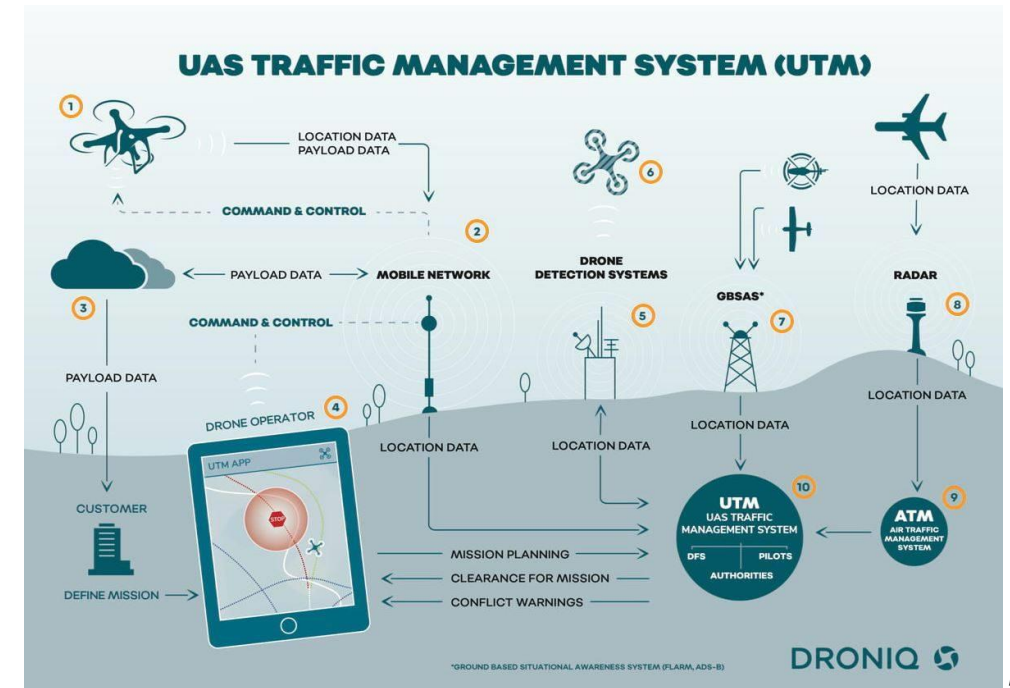
Ericsson, Qualcomm and Dronus demo 5G use case with mmWave-powered drone at Ericsson USA 5G Smart Factory

Available in English 简体中文

A manufacturing and warehousing use case involving a 5G millimeter wave-powered drone has been successfully tested by Ericsson in collaboration with Qualcomm Technologies, Inc. and industrial drone solutions provider Dronus. The 5G mmWave Drone use case is the first step toward production use of Unmanned Aerial Vehicles in a manufacturing environment.

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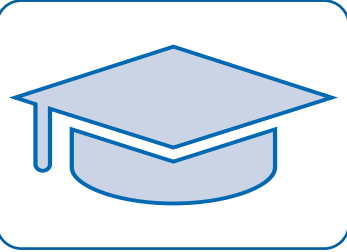
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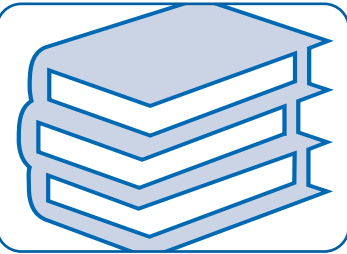
'There can be no AAM industry absent spectrum and communications infrastructure' – Bill Tolpegin Founder AURA Networks Systems [6]

Short Introduction



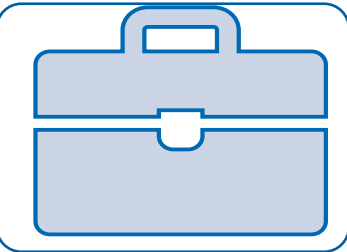
Qualification

- B.Tech Electronics and Communication, SRMIST, India
- M.Sc. Communications Engineering , TU Munich



Relevant Experience

- 2.4 years research assistant at Atmosphere GmBH, Weßling, Munich (Evaluation on Iridium Satellite Communication for UAS)
- Master Thesis: MAVLink Protocol enhancements for UAV Command and Control communication in integrated multi-link scenario (GLOBECOM 2024)



Current position

- PhD in RTG 2947 AirMetro
- Research area : T9 – Wireless Communication Networks @ Deutsche Telekom Chair of Communication Networks

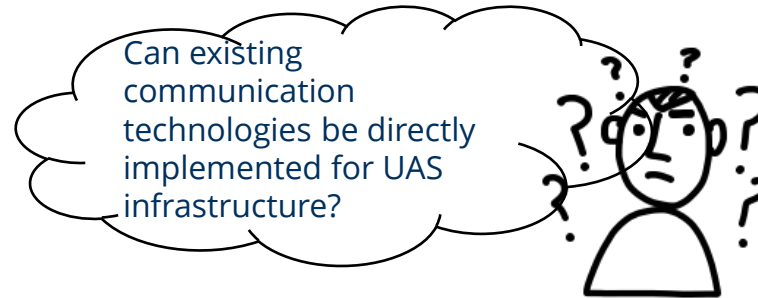


Outlook

- **Introduction and Motivation**
- **Tools for UAS Communication Infrastructure**
- **Proposed Research Methodology**
- **Identified Challenges**
- **Outlook**
- **References**

Introduction and Motivation

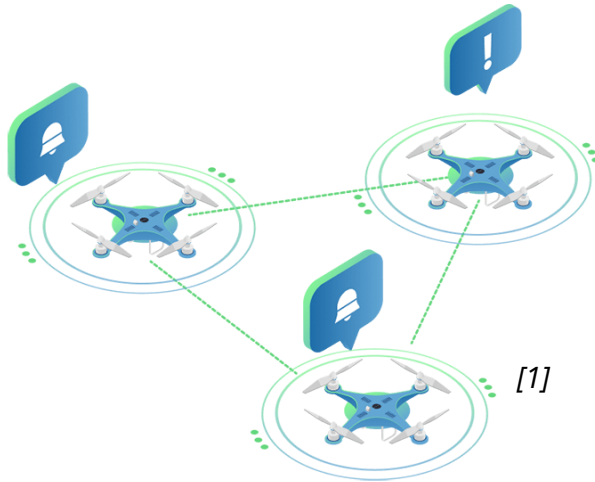
- The conventional use of Communication Technology in Aviation is transformed
- **Terrestrial technology** (5G, 6G) became a crucial part in Advanced Air Mobility infrastructure
- Redundancy is the key
- ICAO recommends **Multi-Link Architecture** for safe operation []
- **Reliable and Resilient Network** (time and error sensitive, capacity demanding) is essential



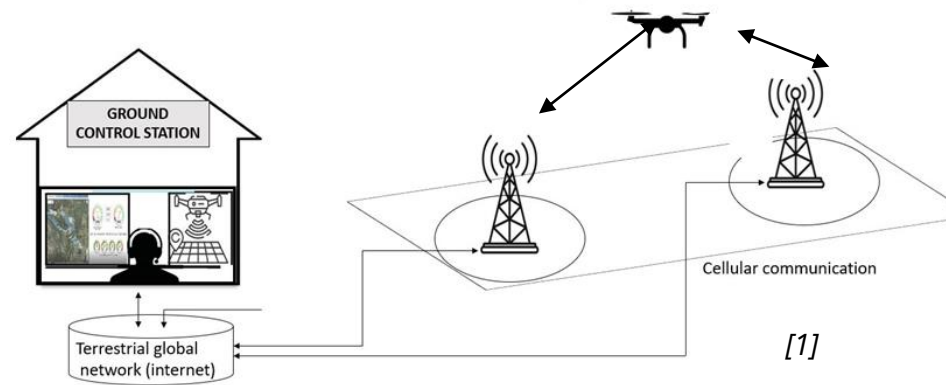
- **Required Communication Performance = Availability, Continuity , Transaction Time, Integrity []**

Tools for UAS communication infrastructure

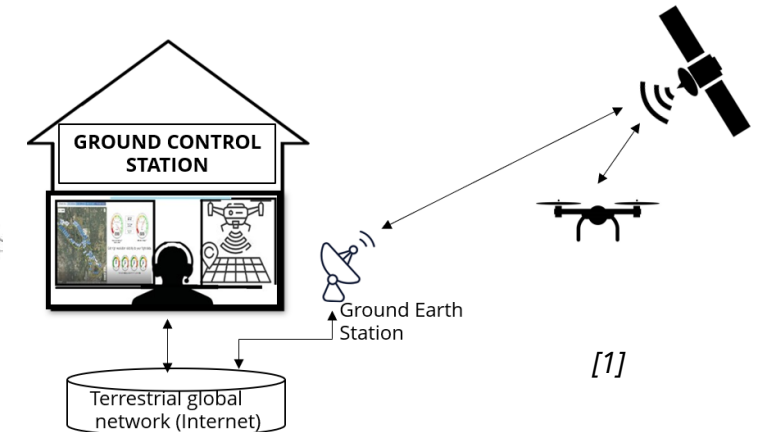
Mesh WLAN Technology



5G Technology

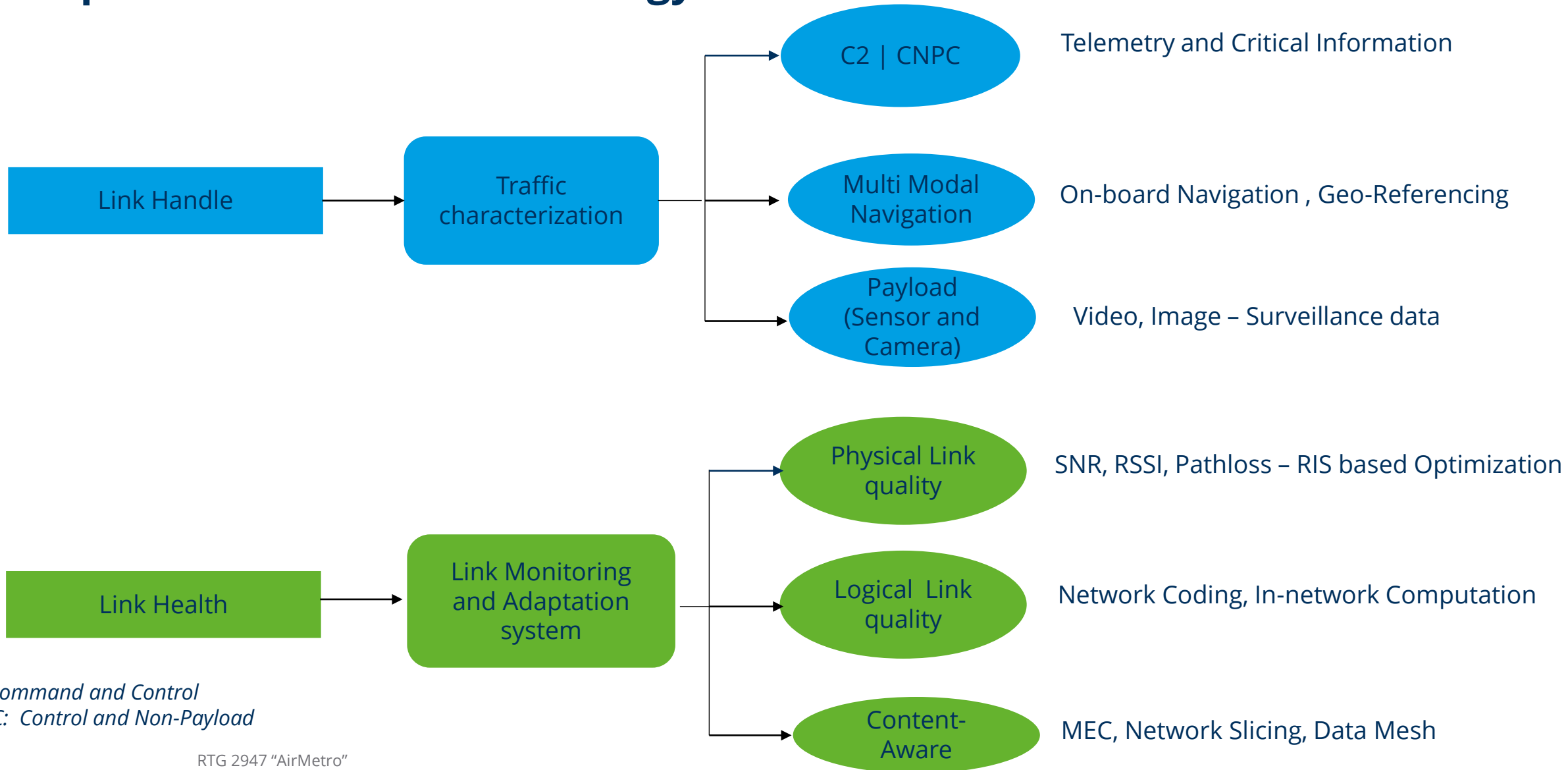


LEO Satellite Technology



1. Link Availability
2. Link Continuity
3. Bandwidth
4. Low Latency
5. Low Packet Error

Proposed Research Methodology



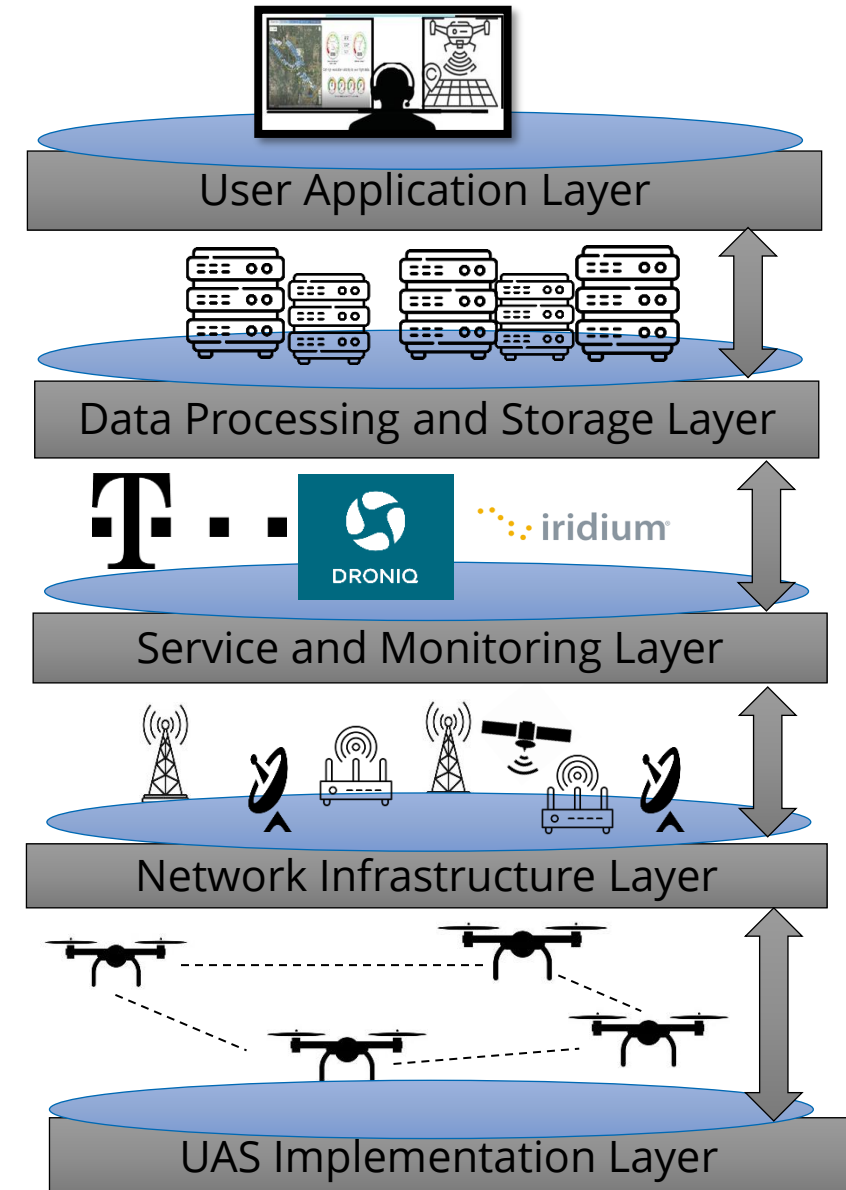
C2: Command and Control
CNPC: Control and Non-Payload

RTG 2947 "AirMetro"
Network Handling and Health for Reliable Communication in AAM
Sridharan, Shreeja
Dresden // 22.11.2024

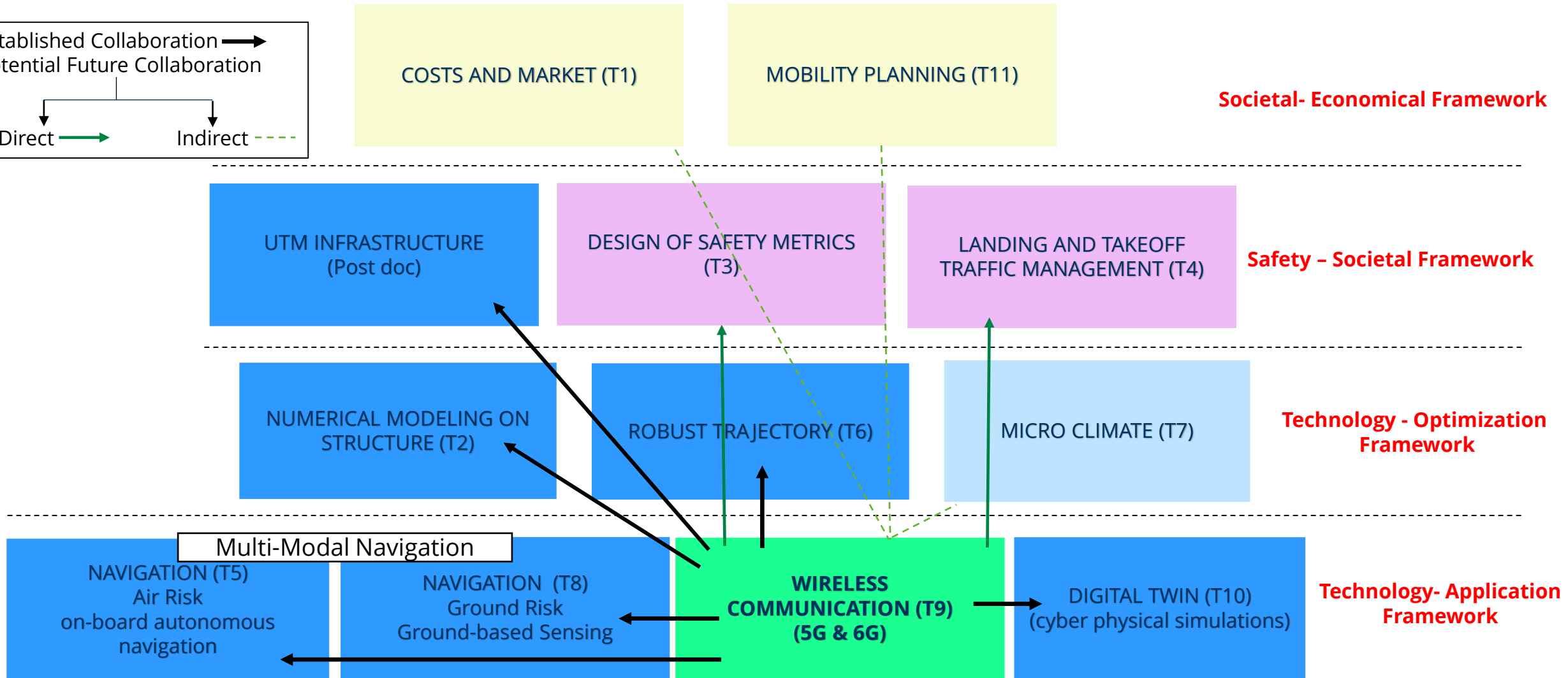
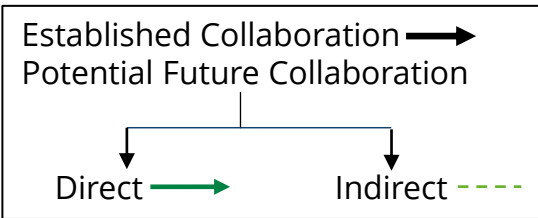
Identified Challenges

- Complexities in **Multi-Link** interfaces (reliable communication architecture)
- Resilient **Network Architecture** - based on CONOPS requirements
- **Data collection** (unrestricted environment for observation)
- Application of various **Communication Link** (Air-to-Ground , Air-to-Air)
- Properties of **Traffic Flow Characterization**
- Convergence of **Physical and Logical link quality indicators** for adaptive implementation

CONOPS: Concept of Operation



Collaboration within RTG AirMetro



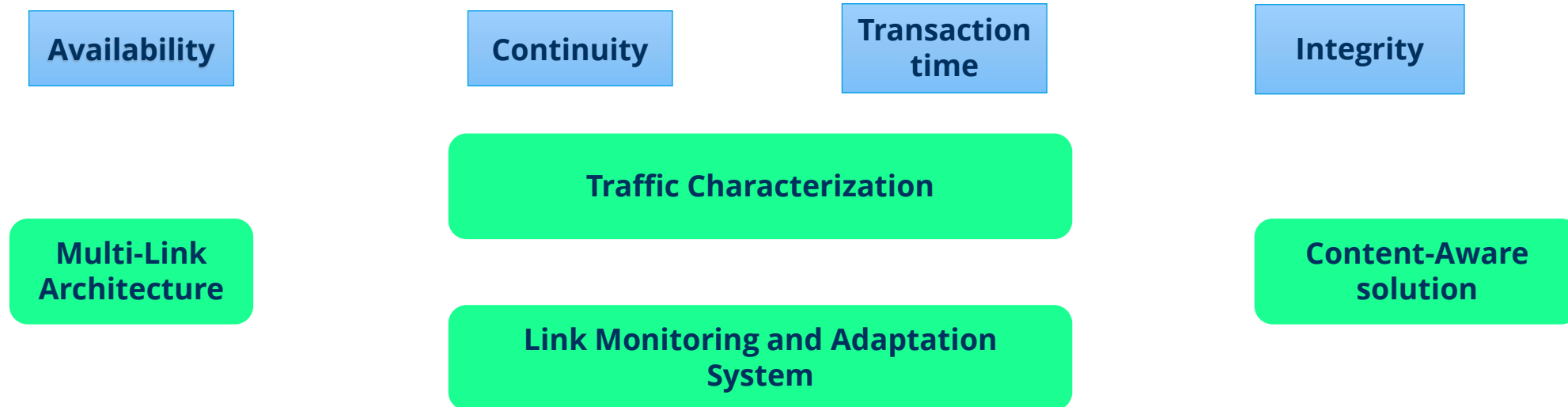
Established Collaboration within RTG AirMetro

Topic Number	Research Domain	Methods	Infrastructure
T2	Numerical Modeling of Smart Structure	Signal Pathloss and RIS-optimization	Link Health + CNS
T5	On-board Autonomous Navigation		CNS
T8	Ground-based sensing	5G based Positioning	CNS
T10	Digital Twin	UAV-swarm	CNS
T6	Robust Trajectories		CNS
Post-Doc	UTM Infrastructure	Communication Channels	CNS

CNS : Communication, navigation, Surveillance

Outlook

- (1) Adept with improvements on the guidelines and compliance requirements by regulatory bodies
- (2) Striving to meet Required Communication Performance



- (3) Future scope on Secure communication infrastructure in Advanced Air Mobility ecosystem